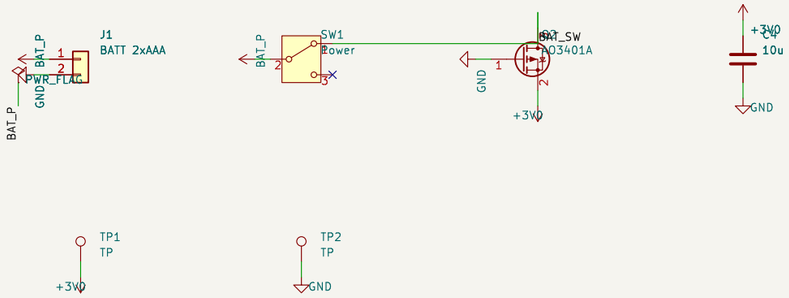
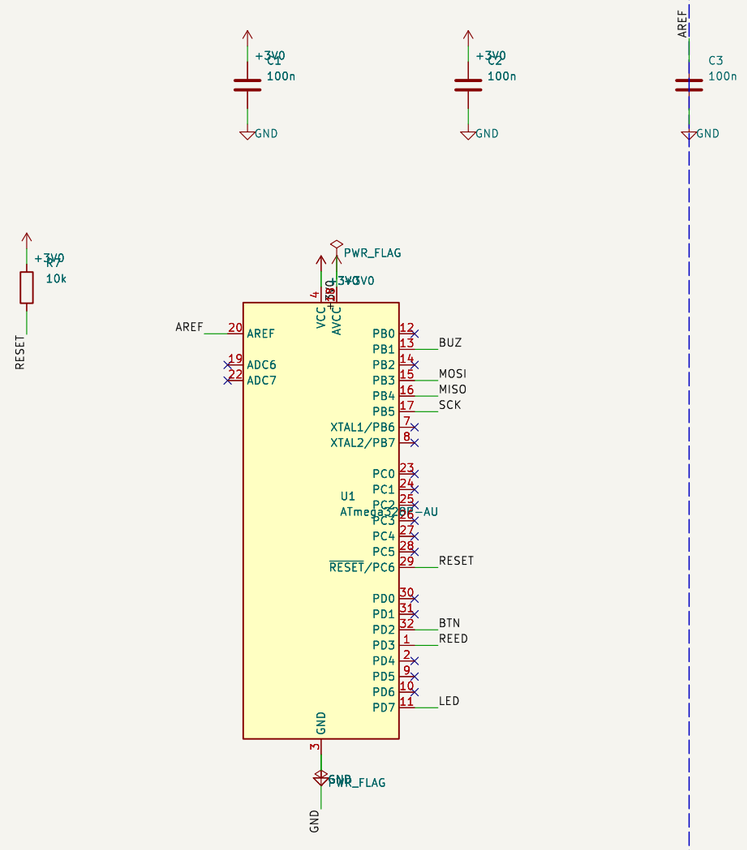
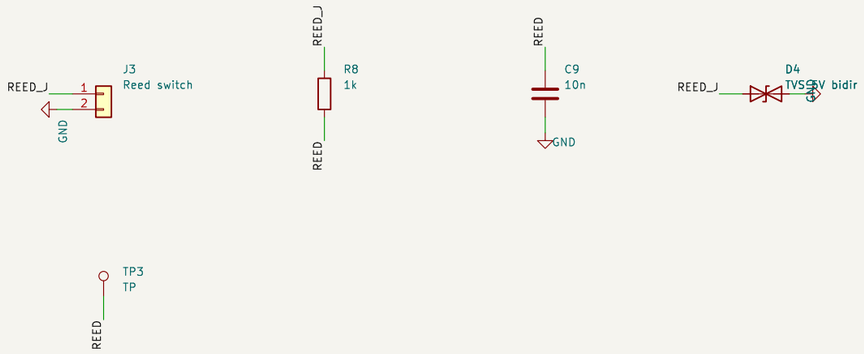
POWER INPUT: 2xAAA, switch, reverse-polarity PMOS



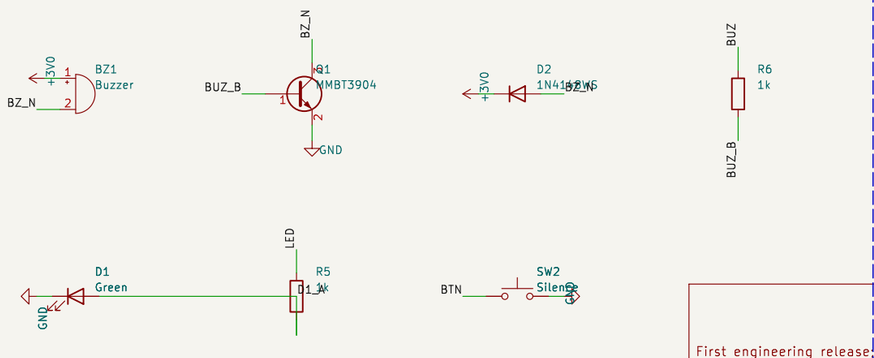
MCU: ATmega328P-AU, 4 MHz, sleeps between door events



REED INPUT (door magnet)



ALARM: buzzer, LED, silence button



Reed switch closes to GND when the magnet is next to it (door shut); door open -> REED goes high -> INT1 wakes the MCU. Internal pull-up on PD3. Sleep current < 2 uA.

Timeline: quiet 50 s, chirp every 10 s, continuous after 180 s; closing the door or the silence button stops it.

PROGRAMMING

First engineering release; see pcb/README.md for status and limits

ClaudeWork

Sheet: /	
File: fridge_door_guard.kicad_sch	
Title: Fridge Door Guard — main board	
Size: A3 — Date: 2026-11-01	**Rev: B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 1/1